

Working Functionalities of SwasthyaSync HMS

Product Name	SwasthyaSync
Version	1.0.0
Document Type	Working Functionalities Reference
Technology Stack	React 18 (Vite) · Node.js (Express) · PostgreSQL
Date	June 2026

1. System Overview

SwasthyaSync is a full-stack, cloud-native Hospital Management System (HMS) designed to digitise and unify all hospital workflows into a single integrated platform. The system covers 40+ hospital modules spanning patient registration, clinical care, diagnostics, pharmacy, billing, revenue cycle, and administration.

System Architecture

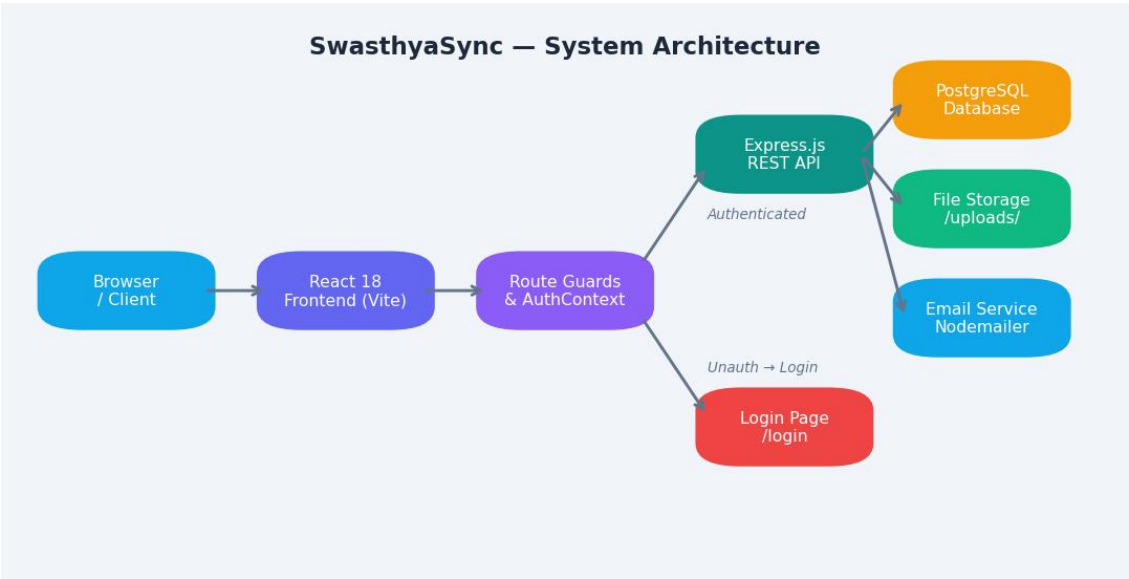


Figure 1.1 — High-Level System Architecture

Key Highlights

Feature	Details
Modules	40+ fully integrated hospital modules
Compliance	NABH documentation standards, Insurance audit trails
Access Control	Role-based access control (Admin / Staff)
Security	JWT-based authentication, bcrypt-encrypted credentials
PDF Generation	Built-in lab reports, discharge summaries, invoices
Form System	PDF annotation engine for hospital consent forms
Deployment	Docker-ready, Vite-built frontend, Express backend

2. Authentication & Access Control

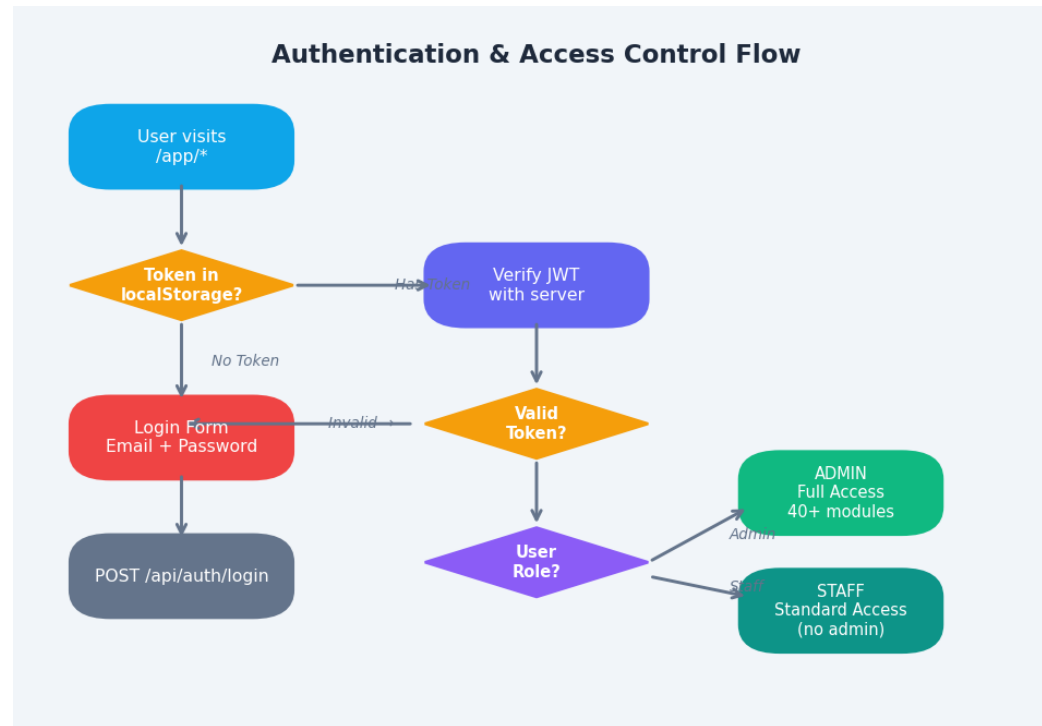


Figure 2.1 — Authentication & Role-Based Access Control Flow

2.1 Login

- Email + password authentication
- JWT tokens issued on login and stored in browser localStorage
- Token verified on every protected route via AuthContext
- Automatic redirect to /login for unauthenticated sessions
- Loading state prevents flash of unauthenticated content

2.2 Registration

- New hospital/user registration form
- Fields: Hospital name, user name, email, password, confirm password
- Password hashing handled server-side (bcrypt)

2.3 Role-Based Access Control

Role	Access Level	Restricted Modules
Admin	Full system access	None — access to all 40+ modules
Staff	Operational access	Staff Management, Form Templates, Audit Log

3. Dashboard & Analytics

The main authenticated landing page provides hospital-wide at-a-glance metrics with live data from all connected modules.

KPI Stat Cards

Metric	Description	Trend
Total Patients	All patients registered in the system	↑ % vs last period
Bed Occupancy	Current % with bed detail (occupied/total)	↑ % change
Pending Notes	Clinical notes awaiting digitisation	↓ (target: reduce)
Today OPD	Outpatient visits today	↑ % change
Revenue (Month)	Monthly billed revenue vs. last month	↑/↓ %
Pending Lab Tests	Lab orders awaiting results	Count
Critical Patients	Patients flagged as critical	Count
Today Revenue	Billing amount generated today	Rs value

Module Launcher — 13 Category Tabs

Category Tab	Key Modules
Patient Administration	Registration, OPD, IPD, Queue, QR, Barcode, Merging, Medical Records
Clinical	Doctors Workbench, Clinical Notes, Discharge Summary, Patient Forms
Patient Billing	OPD Billing, IPD Billing, Payment Collection, GST Management
Revenue Cycle	Claims Processing, Claim Submission, Medical Audit, Alerts
Radiology	Radiology Orders, Report Management, PACS Integration
Laboratory	Lab Orders, Result Entry, Sample Collection, Lab Reports
Nursing Management	Nurse Station, Medication Admin, Vitals Monitoring, Ward Rounds
Operation Theatre	OT Scheduling, OT Notes, Anaesthesia Notes, OT Reports
Blood Bank	Blood Inventory, Donor Management, Blood Requests, Reports
Pharmacy	Drug Dispensing, Drug Inventory, Pharmacy Billing, Prescriptions
Inventory Management	Stock Management, Purchase Orders, Vendor Management
Analytics	MIS Dashboard, MIS Reports, Revenue Analytics, Operational Reports
Emergency	Emergency Triage, Ambulance Management, Critical Care

4. Patient Administration

4.1 Patient Lifecycle

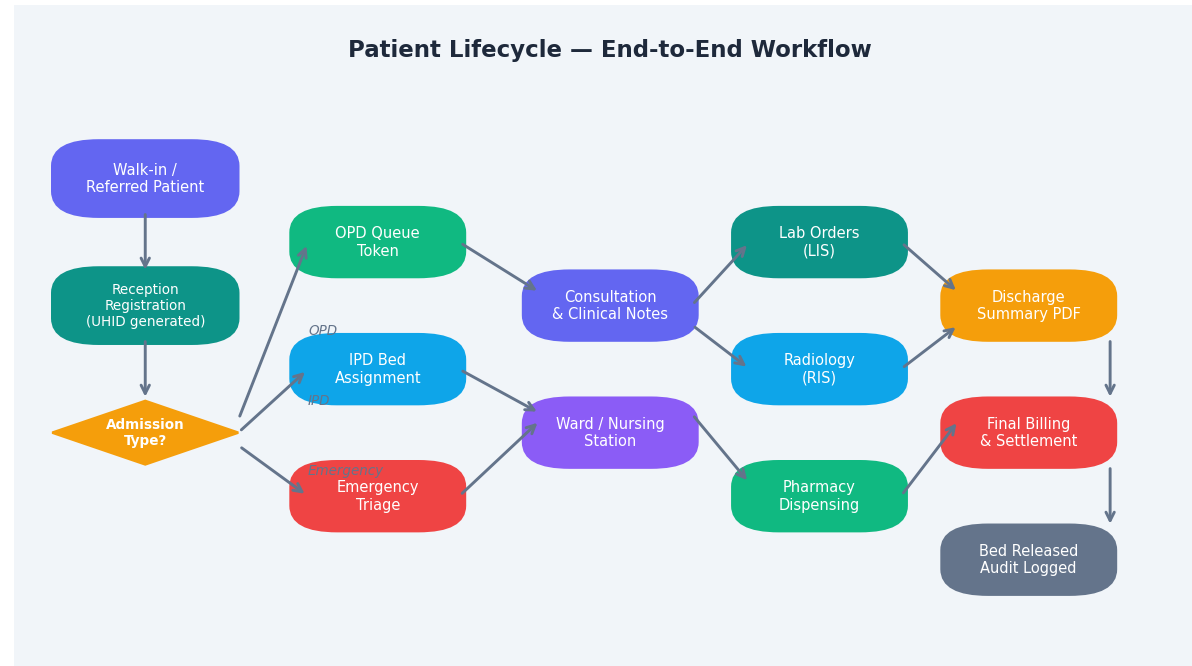


Figure 4.1 — Patient End-to-End Lifecycle

4.2 Patient Registration — 4-Step Wizard

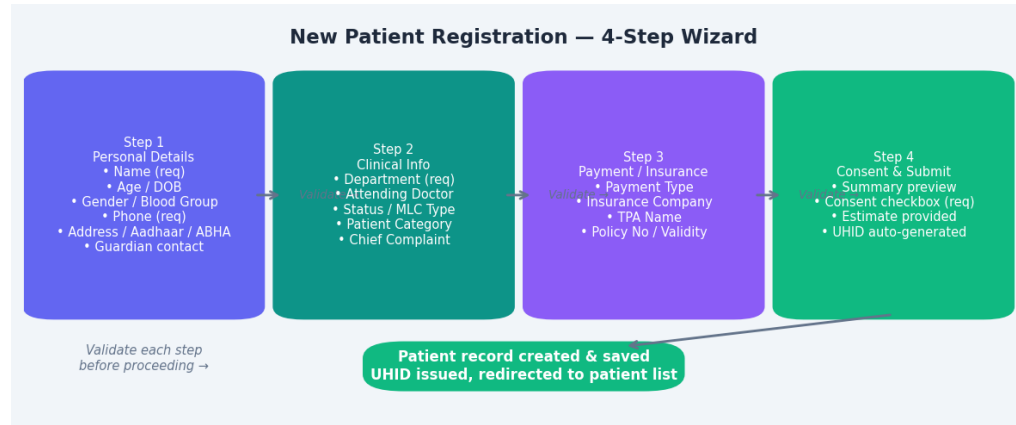


Figure 4.2 — New Patient Registration Wizard Flow

Step 1 Personal Details Name · Age · Gender · Blood Group · Contact · Address · ABHA · Guardian	Step 2 Clinical Info Department · Doctor · Status · MLC Type · Chief Complaint	Step 3 Payment/Insurance Payment Type · Insurance · TPA · Policy No · Validity	Step 4 Consent & Submit Summary review · Consent checkbox · Estimate provided · UHID issued
---	--	--	---

Step 1 — Personal Details Fields

Field	Details	Required
Full Name	Patient full name	Yes
Admission Type	OPD / IPD / Emergency / Day Care / ICU Direct	Yes
Age	Numeric (years)	Yes
Date of Birth	Date picker	No
Gender	Male / Female / Other / Prefer not to say	Yes
Blood Group	A+ / A- / B+ / B- / AB+ / AB- / O+ / O- / Unknown	No
Mobile Number	10-digit mobile	Yes
Aadhaar Number	XXXX XXXX XXXX format	No
ABHA Health ID	14-digit ABHA number	No
Address	Street, city, state (all Indian states), PIN	No
Guardian Name / Relation / Phone	Emergency contact details	No

Step 2 — Clinical Info Fields

Field	Details
Department	30+ departments (General Medicine, ICU, Cardiology, Neurology, etc.)
Attending Doctor	Dropdown from registered doctors
Initial Status	Stable / Critical / Recovering / Under Obs / Serious
Patient Category	General / BPL / Senior Citizen / Divyangjan / VIP / Staff
MLC Type	None / Road Accident / Assault / Poisoning / Burns / Suicide Attempt / etc.
Chief Complaint	Free-text textarea for presenting symptoms

Step 3 — Payment / Insurance Fields

Field	Details
Payment Type	Self Pay (Cash/UPI) / Insurance/TPA / CGHS / ECHS / ESI / Ayushman Bharat / Govt/Free
Insurance Company	Visible only when Insurance/TPA selected
TPA Name	Third-party administrator name
Policy / Member ID	Policy number
Policy Validity	Expiry date picker

4.3 Patient Detail Panel — Tabs

Tab	Contents
Overview	MLC alert banner, key info grid, chief complaint, quick navigation to Forms tab
Clinical	Full demographics, Aadhaar, ABHA ID, insurance details section
Forms	Assigned form instances, assign new template, open FormViewer PDF annotator
Discharge	Assigned discharge summaries, assign template, open DischargeEditor HTML editor

5. Inpatient Department (IPD)

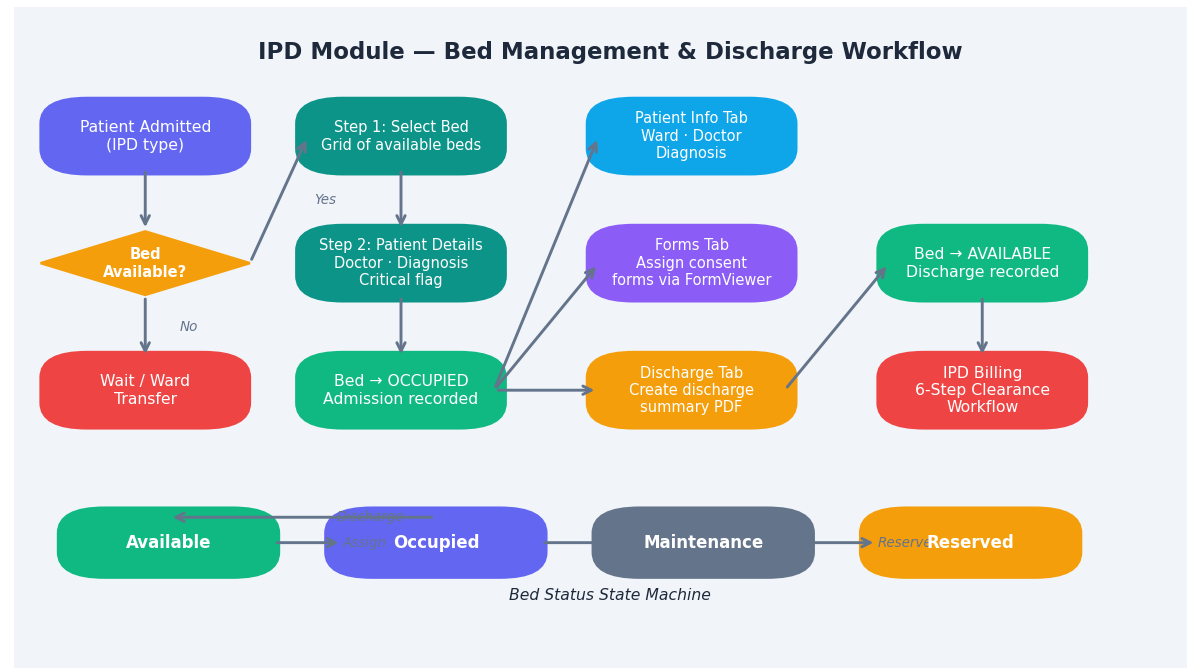


Figure 5.1 — IPD Bed Management & Discharge Workflow

Bed Status Definitions

Status	Colour	Description	Action on Click
Occupied	● Purple	Patient assigned: shows name, age, doctor, diagnosis	Open Bed Detail Panel
Available	● Green	No patient assigned	Open Assign Bed modal
Alert	● Red dot	Pulsing indicator — critical patient	Open Bed Detail Panel
Maintenance	⚙ Gray	Under repair/cleaning	No action
Reserved	● Amber	Reserved for upcoming admission	No action

Bed Detail Panel Tabs

Tab	Contents & Actions
Patient Info	Ward, bed type, attending doctor, admission date, diagnosis. Quick links to Forms and Discharge tabs. Buttons: View Full Record, Write Clinical Note, Generate Discharge Summary, Release Bed
Forms	List of assigned forms with status (Blank/In Progress/Completed). Assign new templates from categorised library. Opens FormViewer on click.
Discharge	Assign discharge summary templates. Opens DischargeEditor (HTML rich text). Download as PDF. Track status.

6. Emergency Module

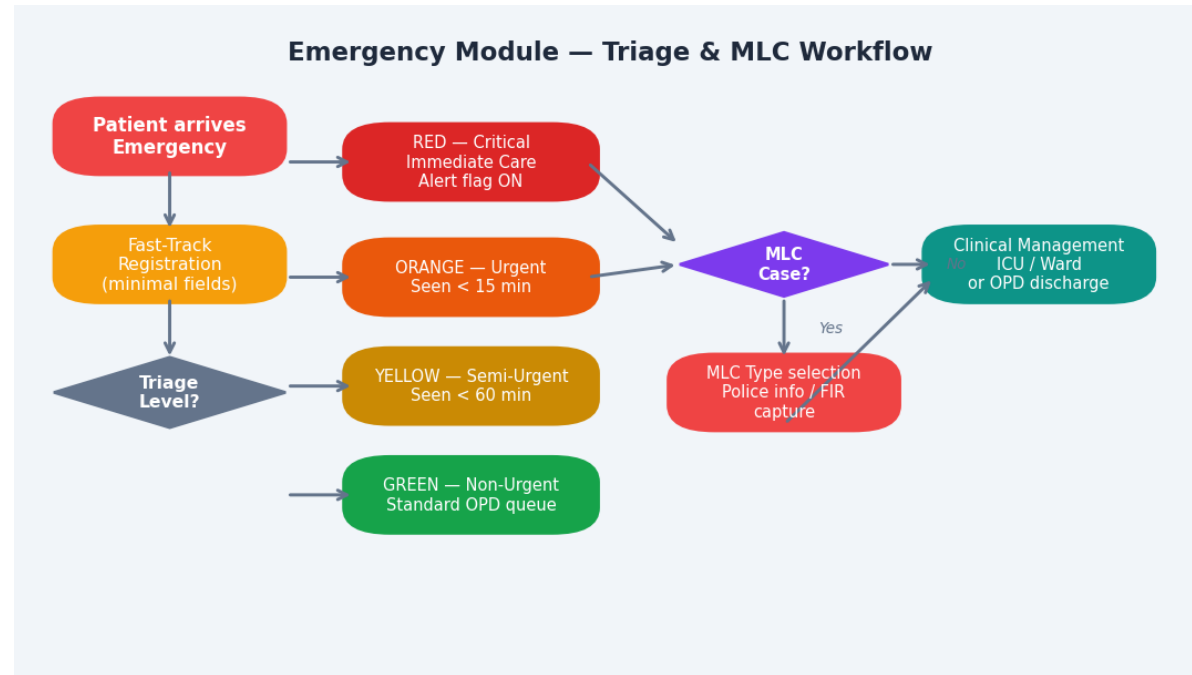


Figure 6.1 — Emergency Triage & MLC Workflow

Triage Levels

Level	Colour	Description	Response Time
Red — Critical	●	Life-threatening. Alert flag ON. Immediate care required.	Immediate
Orange — Urgent	●	Serious condition. Priority care.	Within 15 minutes
Yellow — Semi-Urgent	●	Moderate condition. Can wait briefly.	Within 60 minutes
Green — Non-Urgent	●	Minor condition. Standard OPD queue.	Standard queue

MLC (Medico-Legal Case) Types

MLC Type
Road Traffic Accident (RTA)
Assault
Poisoning
Burns
Sexual Assault
Suicide Attempt
Industrial Accident
Other MLC

7. Appointments & Scheduling

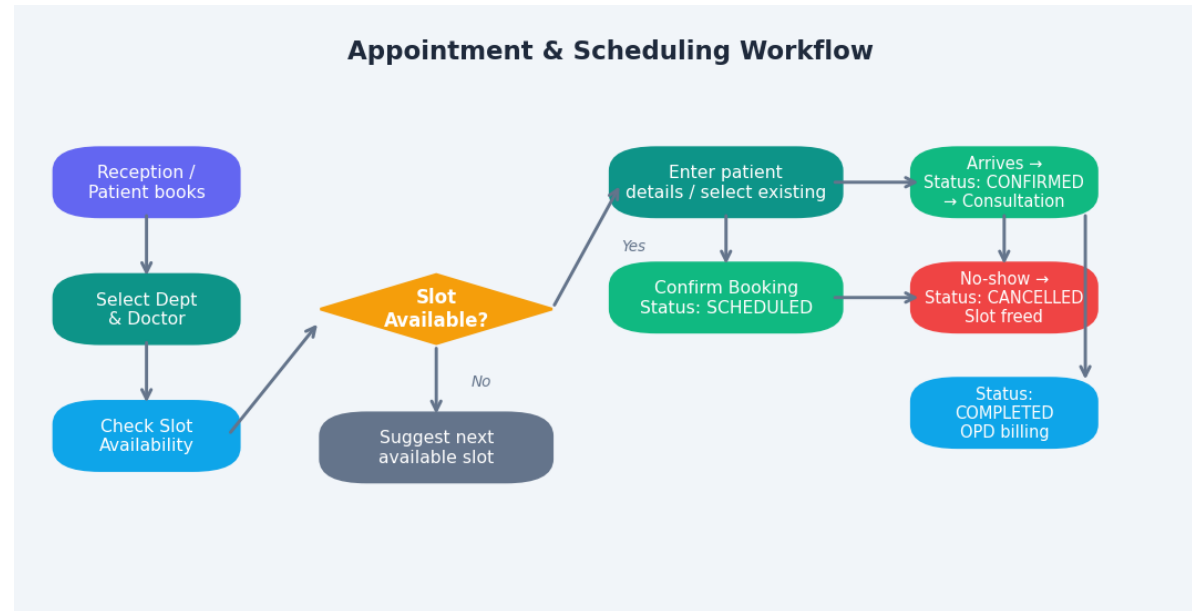


Figure 7.1 — Appointment Booking Workflow

Status	Description
Scheduled	Appointment booked, patient not yet arrived
Confirmed	Patient has arrived and checked in
Completed	Consultation done, OPD workflow begins
Cancelled	Patient did not arrive or appointment was cancelled

8. Clinical Modules

8.1 CPOE — Computerised Physician Order Entry

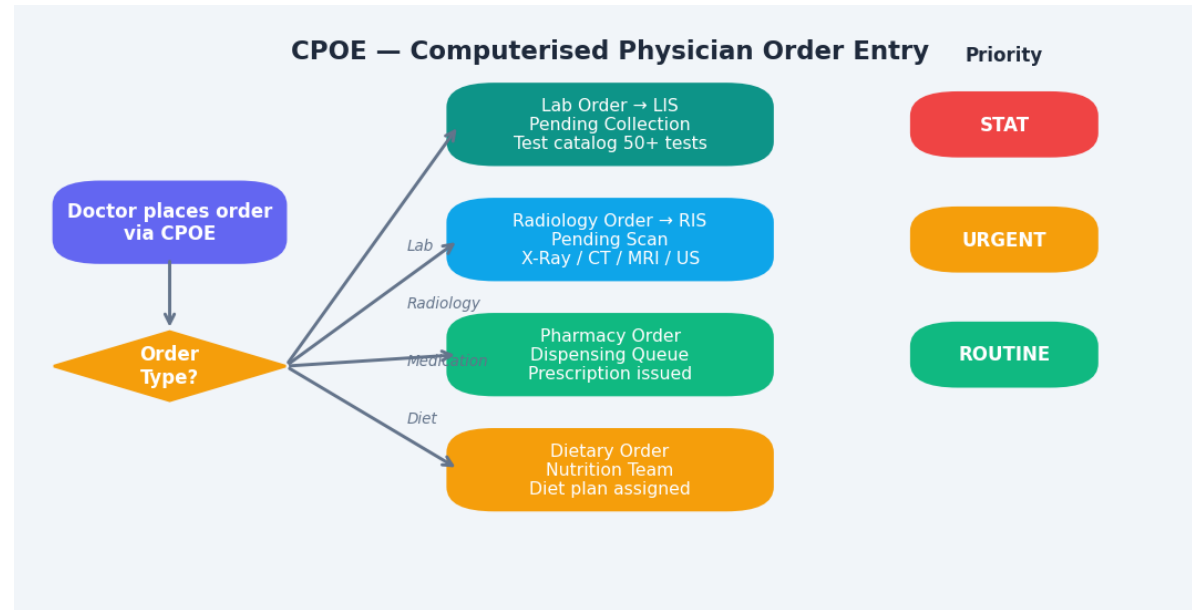


Figure 8.1 — CPOE Order Entry Flow

8.2 Discharge Summary Workflow

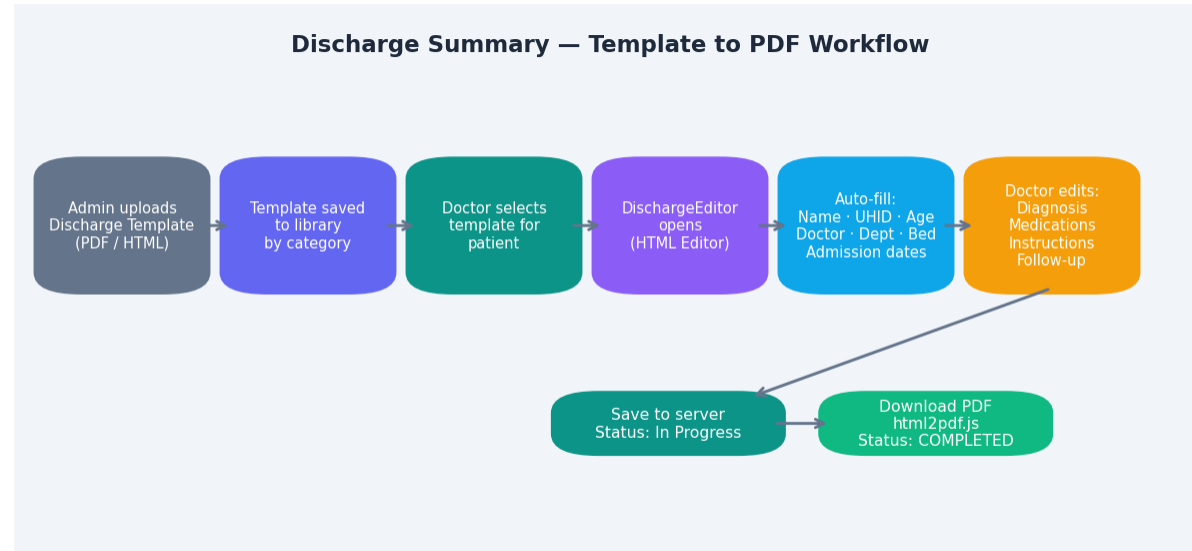


Figure 8.2 — Discharge Summary Template to PDF Workflow

8.3 Patient Forms & Consent Management — FormViewer

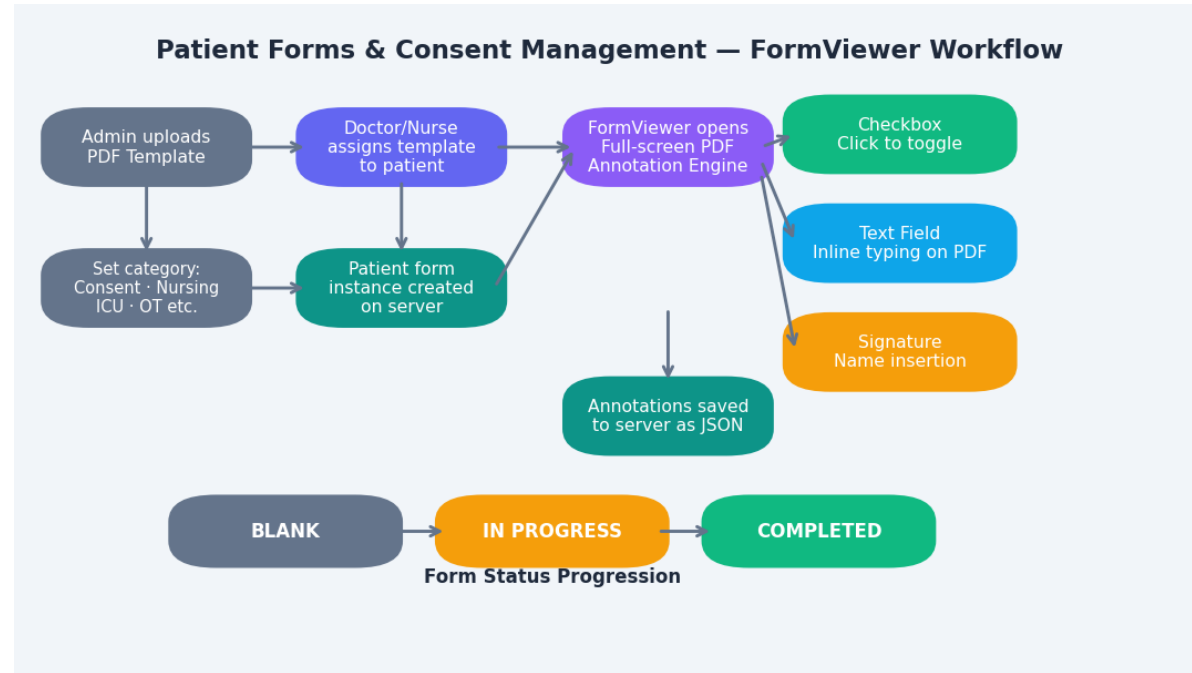


Figure 8.3 — FormViewer PDF Annotation Engine Workflow

Form Categories

Category	Typical Forms
Consent	General Consent for Treatment, Surgical Consent, Anaesthesia Consent
Assessment	Initial Assessment Form, Nursing Assessment, Pain Assessment
Nursing	Nursing Notes, Medication Administration Record (MAR)
ICU	ICU Admission Form, Critical Care Daily Assessment
OT	Pre-operative Checklist, OT Consent, Post-op Record
Emergency	Emergency Assessment, MLC Documentation Form
Discharge	Discharge Instructions, Against Medical Advice (AMA) Form
General	Patient Rights & Responsibilities, Hospital Policy Acknowledgement

Clinical Note Types

Note Type	Description
Admission Note	Initial assessment on patient admission
Progress Note	Daily/shift update on patient condition
Procedure Note	Documentation of a clinical procedure performed
Consultant Note	Opinion from a specialist consultant
Nursing Note	Nursing observations and care provided

9. Laboratory Information System (LIS)

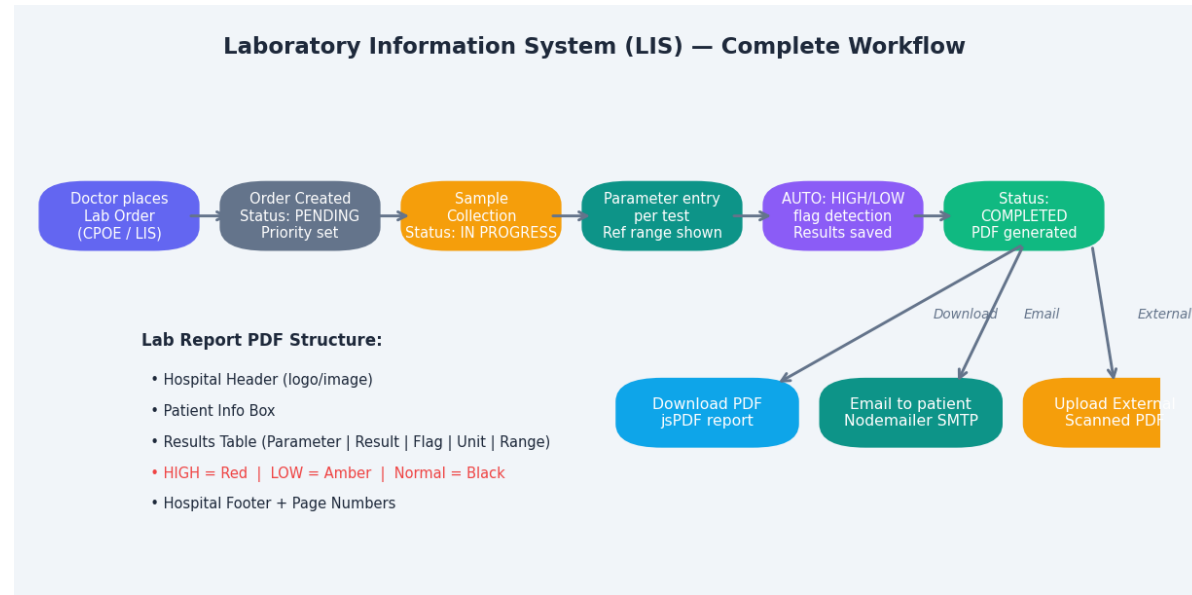


Figure 9.1 — Lab Order to PDF Report Complete Workflow

Lab Test Catalog — 50+ Tests by Category

Category	Tests (examples)
Haematology	CBC, ESR, PT/INR, APTT, D-Dimer, G6PD, Blood Group & Rh
Biochemistry	LFT, KFT, Lipid Profile, Blood Sugar (Fasting/PP/Random), ABG, Amylase, Lipase, Electrolytes
Pathology	Thyroid Profile (T3/T4/TSH), HbA1c, Vitamin D, B12, PSA, CEA, CA125, CA19-9, AFP, Beta HCG
Microbiology	Widal, Dengue NS1/IgG/IgM, Malaria Antigen, HIV, HBsAg, HCV, VDRL, Blood/Urine Cultures, Mantoux
Immunology	CRP, RA Factor, ASO Titre
Cardiology	Troponin I/T, CPK, CPK-MB, LDH

Lab Order Priority Levels

Priority	Description	Turnaround
STAT	Life-threatening emergency — process immediately	Immediate
URGENT	Clinically significant — process as soon as possible	Within 1-2 hours
ROUTINE	Standard testing — normal processing queue	Standard TAT

Lab Report PDF Contents

PDF Section	Contents
Header	Hospital logo + name + address + tagline (Text/Logo mode) OR full header image (Image mode)
Report Title	LABORATORY TEST REPORT
Patient Info Box	Patient Name, Patient ID, Requesting Doctor, Report Date & Time
Investigation Title	Test name (e.g. CBC, LFT, etc.)
Results Table	Parameter Result Flag (HIGH/LOW) Unit Biological Reference Range
Flag Indicators	HIGH values in Red, LOW values in Amber, Normal values in Black
Footer	Lab director signature, accreditation info, page numbers (Text mode) OR full footer image

10. Billing & Revenue Cycle

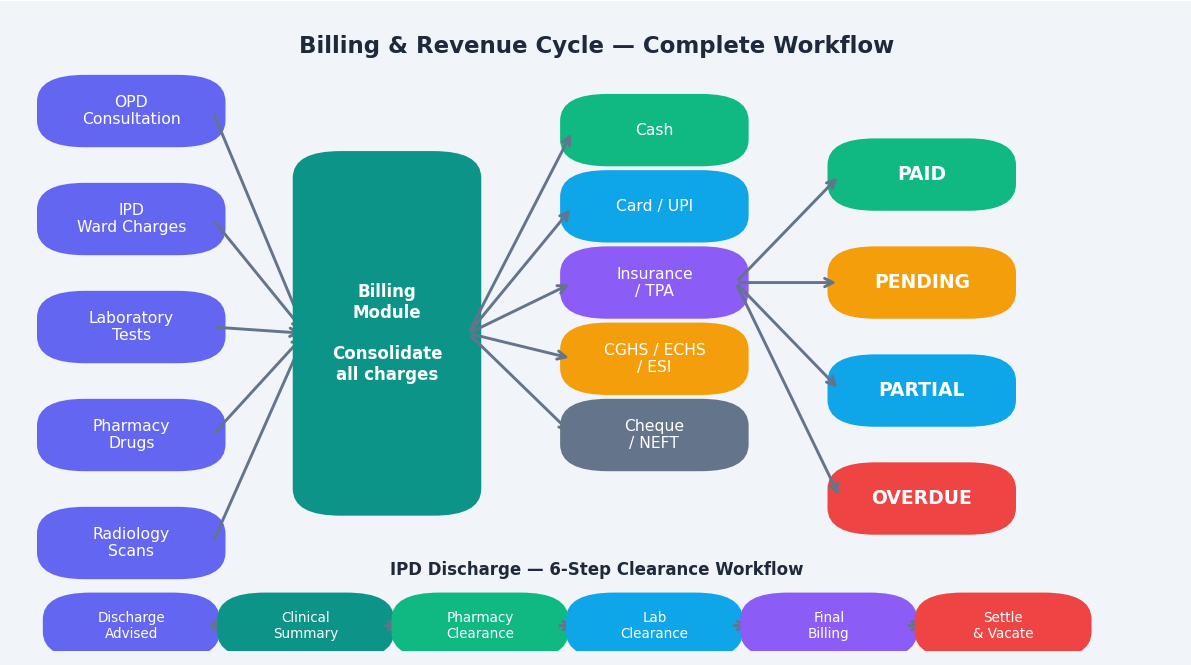


Figure 10.1 — Complete Billing & Revenue Cycle Workflow

Invoice Status Types

Status	Description
Paid 	Full payment collected. Bill settled.
Pending 	Invoice generated, payment not yet received.
Partial 	Partial payment received, balance outstanding.
Overdue 	Payment due date passed, not yet settled.

Accepted Payment Methods

Method	Description
Cash	Cash payment at counter

Card	Debit/Credit card payment
UPI	UPI payment (GPay, PhonePe, Paytm etc.)
Insurance	Third-party insurance settlement
Cheque	Cheque payment
NEFT	Bank transfer via NEFT/RTGS

IPD Discharge Billing — 6-Step Clearance Workflow

Step	Stage	Description
0	Discharge Advised	Doctor advises discharge. Process initiated.
1	Clinical Summary	Doctor completes clinical summary and signs off on case.
2	Pharmacy Clearance	All pharmacy charges verified and added to bill.
3	Laboratory Clearance	All lab test charges verified and added to bill.
4	Final Billing	All charges consolidated: ward, nursing, OT, lab, pharmacy, radiology.
5	Settle & Vacate	Patient pays final bill. Bed released. Admission closed.

TPA / Insurance Management

- Insurance company and TPA master data management
- Pre-authorisation request tracking and submission
- Claims processing and submission to insurance company
- Medical audit trail — full documentation for insurance review
- Technical audit documentation for claim processing
- Alerts management for claim status updates and queries

11. Settings & Hospital Configuration

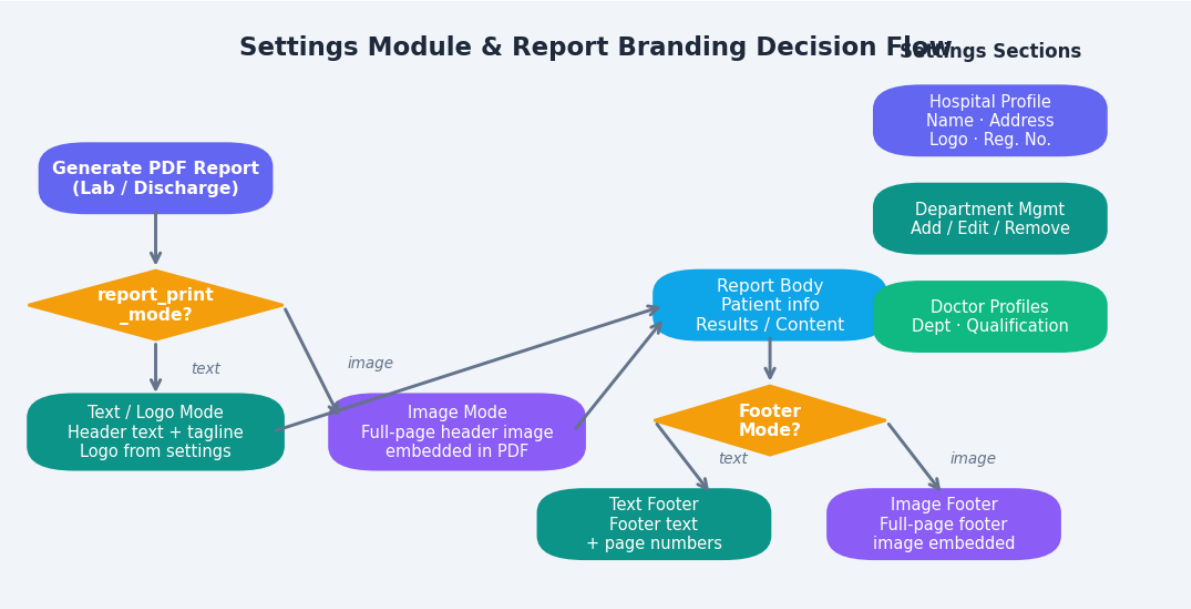


Figure 11.1 — Settings Module & Report Branding Decision Flow

Report Print Modes

Mode	Header	Footer	Use Case
Text / Logo	Hospital name + tagline + logo from settings	Text footer + page numbers	Standard branded reports
Image Mode	Full-page header image (uploaded by admin)	Full-page footer image (uploaded by admin)	Pre-designed letterhead/stationery

12. Other Clinical & Operational Modules

12.1 Radiology Information System (RIS)

Feature	Details
Order Creation	Radiology order linked to patient and requesting doctor
Investigation Types	X-Ray, CT Scan, MRI, Ultrasound, ECG, Echo, and more
PACS Integration	Integration point for PACS (Picture Archiving and Communication System)
Report Management	Type report in system or upload scanned PDF
Status Tracking	Pending → Scanned → Reported

12.2 Pharmacy Management

Feature	Details
Drug Dispensing	Prescription-based drug dispensing from doctor orders
Drug Inventory	Stock tracking with low-stock alerts
Pharmacy Billing	Drug charges automatically added to patient bill
Drug Catalog	Medicine master with dosage and category

12.3 Blood Bank

Feature	Details
Inventory Management	Blood stock tracking by blood group (A+/A-/B+/B-/AB+/AB-/O+/O-)
Donor Management	Donor registration, donation history, eligibility check
Blood Components	Whole Blood, Packed RBCs, Platelets, FFP, Cryoprecipitate
Blood Requests	Request linked to patient, compatibility verification

12.4 Operation Theatre (OT)

Stage	Documentation
Scheduling	OT booking: date, time, OT room, surgeon, anaesthesiologist
Pre-operative	Surgery checklist, pre-op notes, patient briefing
Anaesthesia	Anaesthesia notes by anaesthesiologist
Intra-operative	OT notes: surgical findings, procedure steps
Post-operative	Recovery room monitoring, post-op instructions

12.5 Inventory Management

- Stock item master: medicines, consumables, medical equipment

- Purchase order creation and vendor tracking
- Goods receipt and automatic stock increment on receipt
- Stock level monitoring with configurable low-stock alerts
- Vendor master management with contact and payment terms
- Reports: current stock, stock movement, inventory valuation

12.6 Staff Management (Admin Only)

- Staff registration: doctors, nurses, administrative staff, lab technicians
- Role assignment: Admin or Staff
- Department allocation and designation management
- Active / Inactive status control
- Staff list with search and filter

13. Reports & MIS

Report Type	Contents
MIS Dashboard	Hospital-level KPI overview with live metrics
MIS Reports	Exportable data: admissions, discharges, OPD visits
Revenue Analytics	Revenue trend, department-wise collections, insurance vs. cash split
Operational Reports	Bed occupancy, average length of stay, OT utilisation, lab TAT
Clinical Audit	Quality metrics, clinical outcome tracking, NABH documentation

14. Audit Log (Admin Only)

Complete chronological log of all system actions, tamper-proof for compliance and investigation purposes.

Event Type	Details Logged
Patient created / updated	UHID, fields changed, user, timestamp
Bed assigned / released	Bed ID, patient, doctor, action, timestamp
Lab order / result entered	Order ID, test name, result, user, timestamp
Bill generated / payment received	Invoice ID, amount, payment method, user
Form assigned / completed	Form ID, patient, template, status change
User login / logout	Username, IP, timestamp

Settings changed

Field changed, old value, new value, user

15. Technical Architecture

Frontend Technology Stack

Component	Technology	Purpose
Framework	React 18 with Vite	UI framework with fast HMR development
Routing	React Router v6	Client-side routing with protected routes
Charts	Recharts	Area, Bar, and Pie charts for analytics
Icons	Lucide React	Consistent icon library
PDF Generation	jsPDF + jsPDF-AutoTable + html2pdf.js	Lab reports, discharge summaries, invoices
PDF Rendering	pdfjs-dist	Render PDF forms in browser
QR Code	html5-qrcode	QR code scanning for patient registration
State Management	React Context (AuthContext)	Global auth state management
Styling	Vanilla CSS + CSS custom properties	Design system with tokens

Backend Technology Stack

Component	Technology	Purpose
Runtime	Node.js	JavaScript server runtime
Framework	Express.js	REST API server
Database	PostgreSQL	Relational data storage
Authentication	JWT (jsonwebtoken) + bcrypt	Token auth + password hashing
File Storage	Local filesystem /server/uploads/	PDFs, images, logos
Email	Nodemailer (SMTP)	Lab report email delivery

Key Database Tables

Table	Purpose
patients	Master patient record with all demographics, clinical info, payment details
beds	IPD bed registry with status, ward, patient assignment
lab_orders	Lab test orders, results (JSON), PDF path, status
radiology_orders	Radiology investigation orders and reports
billing	Invoice and payment records with status tracking
appointments	OPD appointment bookings with slot management
doctors	Doctor master: name, department, qualification
clinical_notes	Doctor clinical documentation with priority and type
form_templates	Uploaded PDF form templates with category
patient_forms	Patient-specific form instances with annotations JSON

discharge_templates	Discharge summary template library
discharge_summaries	Patient-specific discharge summary instances
staff	Staff and user records with role and department
audit_log	Tamper-proof system action audit trail
hospital_settings	Hospital profile, branding, report configuration

Deployment Configuration

Component	Details
Containerisation	Docker + Docker Compose
Frontend build	npm run build (Vite production bundle)
Dev server	npm run dev (Vite with HMR on port 5173)
Backend	Node.js Express server (port configurable via .env)
Database	PostgreSQL (Docker container or managed instance)
Environment	.env file for DB credentials, JWT secret, SMTP config

*Document prepared from codebase analysis of the SwasthyaSync HMS project.
© 2026 SwasthyaSync. All rights reserved.*